

significant power, making edge AI essential. By processing data locally, devices maintain efficient operation even without constant connectivity.

In the medical sphere, AI helps doctors make quicker decisions. MRI and CT scans produce massive amounts of information for humans to review. With AI, patterns can be quickly identified, issues can be detected, and possible diagnoses suggested, enabling faster patient care without relying solely on specialists.

Smarter connectivity: The 5G revolution

The main core of any embedded system is connectivity, especially the demanding and fast-growing

Connectivity is the heartbeat of modern embedded systems, especially in the fast-growing Internet of Things (IoT) ecosystem. In these early days, communication between devices relied on technologies such as Bluetooth, Wi-Fi, Zigbee, and Thread, which formed the basis of low-power, short-range networking. In turn, enabling various devices to communicate effectively developed the bedrock on which today's connected world was built.

With the full release of 5G, the landscape of how devices communicate has changed. Ultra-fast data speeds, latency at an all-time low, and highly stable connections—all because of 5G-push, the envelope of embedded systems. More defined decision-making and unified automation can now be enabled by devices that communicate in real time with high accuracy and responsiveness.

This surge in connectivity equips a wide range of applications, from wearable health trackers that continuously share vital data to the latest industrial systems, which depend on real-time monitoring and predictive maintenance. Remote sensors, autonomous machines, and intelligent



RedCap

infrastructure will all benefit from dependable, high-speed communication enabled by 5G.

5G is much more than just enhanced connectivity; it reinvents embedded systems into a smart, dynamic network that adapts, learns, and responds in real time to its surroundings.

Beyond faster downloads, 5G introduces critical advancements. It reduces latency for real-time communication, enhances reliability for safety-critical operations, and improves energy efficiency through optimised standards such as RedCap (reduced-capability 5G), making it suitable for low-power IoT devices.

These are benefits that already transform industries. Hyundai replaced its factory Wi-Fi with a private 5G network powered by RedCap, allowing production lines to communicate more quickly and function more smoothly while assembling several thousand cars daily. A move like that—a reduction of interference and improvement in reliability—helps the engineers monitor the equipment in real time and make instantaneous adjustments in manufacturing processes.

5G is enabling entirely new

possibilities. Autonomous drones can navigate urban airspace, and surgeons can perform remote procedures with precision. Beyond communication, 5G facilitates real-time interaction with machines, supporting high-precision operations with minimal latency. Each advancement in 5G technology expands the boundaries of what is possible, enabling systems that are more responsive, capable, and intelligent than ever before.

When machines get smarter

The automotive industry stands at an inflection point where smart technology and connectivity converge to create intelligent, sophisticated vehicles. Today, nearly every vehicle comes equipped with advanced driver-assistance systems (ADAS). Autonomous car trials have also become increasingly common, including in India. AI now extends far beyond navigation, transforming how vehicles operate, maintain themselves, and respond to their environments.

Now, AI has moved beyond the cloud and is increasingly integrated into vehicles' core systems. AI provides early warning through many new sensor technologies. It can also schedule maintenance to prevent breakdowns and what not. AI works basically on data collected from each corner of the vehicle. Faults can be predicted before they occur using data, thereby optimising engine performance.

Challenges are something that come with everything. Connectivity is not spared, either, as it has its own set of new challenges. Modern vehicles are full of features that are very appealing to cybercriminals. Engineers need to minimise this threat by adding security at the hardware level. Now the question is, how can engineers do this? They can use secure boot processes, encryption-enabled processors, and attestation protocols to ensure that all components running in a vehicle